Yashraj Singh Thakur

+91 7000501987 Mail LinkedIn Github Portfolio Leetcode

Education

Narmada Valley International School, Narmadapuram **2019 – 2020**
Class 10th *92.8%*

Narmada Valley International School, Narmadapuram **2021 – 2022**
Class 12th *97.4%*

Indian Institute of Technology, Patna **2023 – 2027**
B.Tech in Engineering Physics *8.36/10 CGPA*

Work Experience

AlgoUniversity (backed by Y-Combinator) **May 2025 – June 2025**
SWE Co-Op Intern *Remote*

- Built a code evaluation system supporting **3+ languages** and **100+ concurrent users**.
- Analyzed logs from **5,000+ submissions** and reduced latency by **25%**.
- Implemented authentication for **40+ users** with progress tracking.
- Standardized runtime using Docker to improve deployment stability.

Projects

Cohort – Online Competitive Programming Judge GitHub Deployment

- Handled **1,000+ submissions** using queue-based task processing.
- Monitored traffic from **500+ users** and improved throughput by **20%**.
- Deployed cloud services with **99.9% uptime**.
- Reduced manual effort by **70%** through workflow automation.
- **Tools:** Node.js, MongoDB, Redis, AWS EC2, Vercel.

Algorithmic Trading Strategy GitHub

- Analyzed **10,000+ price points** to generate trade signals.
- Backtested strategy across **3 market conditions**.
- Containerized services for reproducible deployment.
- Measured reliability using **82% test coverage**.

Smart Energy Forecast Dashboard *(FastAPI, ML, RL, PostgreSQL)* GitHub Link

- Trained models on **50,000+ records** to predict energy demand.
- Reduced estimated cost by **15–20%** through scheduling.
- Tracked performance using dashboards with **10+ metrics**.
- Improved resource use using reinforcement learning.
- **Tech:** FastAPI, XGBoost, Prophet, PostgreSQL, Chart.js.

Technical Skills

Programming: Python, Java, JavaScript, C

Data & Systems: SQL, PostgreSQL, MongoDB, Redis, Docker, AWS, GCP

Frameworks: Node.js, Express.js, FastAPI, React

Tools: GitHub, Postman, VS Code